# Brainstorming & Idea Prioritization

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| **Date** | July 9, 2026 |
| **Team ID** | SWTID-2026-1262 |
| **Project Name** | OptiCrop: Smart Agricultural Production Optimization Engine |
| **Maximum Marks** | 3 Marks |

### Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

| S.No | Team Member | Idea / Suggestion | Category | Group No. |
|---|---|---|---|---|
| 1 | Bokam Hari Charan | Real-time predictive crop matching engine utilizing N, P, K soil profiles and live weather data. | Machine Learning Frontend | Group 1 |
| 2 | Bokam Hari Charan | Core algorithmic evaluation loop supporting multi-model pipelines (KNN, Decision Trees, Random Forest, Logistic Regression). | Machine Learning Backend | Group 1 |
| 3 | Bokam Hari Charan | A comparative target crop suitability audit module enabling users to check compatibility metrics for a chosen specific crop. | Feature Module | Group 2 |
| 4 | Bokam Hari Charan | High-level macro analytics panel mapping agro-ecological zoning constraints and optimization roadmaps for regional planning.Policy & Analytics | Policy & Analytics Module | Group 3 |

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| 5 | Bokam Hari Charan | Multi-platform interactive user portal engineered with Flask and structured using Bootstrap responsive layouts. | System Architecture | Group 1 |
| 6 | Bokam Hari Charan | Static edge calculation rule logic strictly based on standard global crop baseline averages. | Alternative Engineering | Group 4 |

**Step 2: Idea Prioritization**

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

| Group No. | Final Idea | Feasibility (High/Medium/Low) | Importance (High/Medium/Low) | Priority | Selected (Yes/No) |
|---|---|---|---|---|---|
| 1 | Build an AI-driven pipeline using an optimized Random Forest classifier integrated into a modular web form mapping real-time Nitrogen, Phosphorus, Potassium, temperature, pH, humidity, and rainfall arrays. | High | High | 1 | Yes |
| 2 | Develop an interactive audit portal where users input soil conditions alongside a target crop to | High | High | 2 | Yes |

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| | verify environmental matching metrics and catch crop-soil variance anomalies. | | | | |
| 3 | Provide a dynamic analytics output rendering specialized macro recommendations detailing environmental resource efficiency roadmaps for regional resource planning. | High | High | 3 | Yes |
| 4 | Scraping a hard-coded look-up table without using machine learning algorithms or pickle persistence files. | High | Low | 4 | No |